SQL Practice Task – Subqueries (Part 1)

Topics Covered

- Scalar Subqueries
- Multi-row Subqueries
- Correlated Subqueries

Instructions

- Create the database and tables as instructed.
- Insert your own sample data (minimum 10 records in each table unless specified).
- Solve every question **using only Subqueries**.
- **Do NOT use JOINs**.
- Execute every query and verify the output.

Database Scenario

You are developing a database for a company called **TechNova Solutions**.

The company has two tables:

- Departments
- Employees

Employees belong to different departments, and the management wants various analytical reports using **Subqueries**.

Create the Tables

Create the following tables:

departments

Column	Data Type
id	INT
department_name	VARCHAR(50)
location	VARCHAR(50)

create table departments(

id int primary key,

department_name varchar(50),

location varchar(50));

insert into departments values

(1,'Engineering','Hyderabad'),

(2,'Sales','Bangalore'),

(3,'HR','Hyderabad'),

(4,'Finance','Chennai')

;

---

employees

Column	Data Type
id	INT
name	VARCHAR(50)
salary	DECIMAL(10,2)
department_id	INT

Use appropriate **PRIMARY KEY** and **FOREIGN KEY** constraints.

Insert:

- At least **4 Departments**
- At least **10 Employees**

Make sure:

- Multiple employees belong to the same department.
- Different departments have different average salaries.
- At least two departments are in the same city.

create table employees(

id int primary key,

name varchar(50),

salary decimal(10,2),

department_id int,

foreign key(department_id) references departments(id)

);

insert into employees values

(101,'Dinesh',80000,1),

(102,'Akhila',68000,2),

(103,'Sushma',65000,1),

(104,'Anitha',75000,2),

(105,'Kalyan',85000,3),

(106,'Babu',35000,4),

(107,'Vignesh',40000,4),

(108,'Siddhu',81000,2),

(109,'Bob',90000,1),

(110,'Pavan',82000,3)

;

Section A – Scalar Subqueries

Use **only Scalar Subqueries**.

1.

Display employees earning more than the company average salary.

```sql
select* from employees
where
salary > (select avg(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Expor

id	name	salary	department_id
101	Dinesh	80000.00	1
104	Anitha	75000.00	2
105	Kalyan	85000.00	3
108	Siddhu	81000.00	2
109	Bob	90000.00	1
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

2.

Display the employee(s) earning the highest salary.

```sql
select * from employees
where
salary = (select max(salary) from employees);
```

id	name	salary	department_id
109	Bob	90000.00	1
NULL	NULL	NULL	NULL

3.

Display the employee(s) earning the lowest salary.

```sql
select * from employees
where
salary = (select min(salary) from employees);
```

id	name	salary	department_id
106	Babu	35000.00	4
NULL	NULL	NULL	NULL

4.

Display employees earning more than the minimum salary.

```sql
select * from employees
where
salary > (select min(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Exp

id	name	salary	department_id
101	Dinesh	80000.00	1
102	Akhila	68000.00	2
103	Sushma	65000.00	1
104	Anitha	75000.00	2
105	Kalyan	85000.00	3
107	Vignesh	40000.00	4
108	Siddhu	81000.00	2
109	Bob	90000.00	1
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

5.

Display employees earning less than the highest salary.

```sql
select * from employees
where
salary < (select max(salary) from employees);
```

Result Grid | Filter Rows: | Edit:

id	name	salary	department_id
101	Dinesh	80000.00	1
102	Akhila	68000.00	2
103	Sushma	65000.00	1
104	Anitha	75000.00	2
105	Kalyan	85000.00	3
106	Babu	35000.00	4
107	Vignesh	40000.00	4
108	Siddhu	81000.00	2
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

6.

Display employees earning exactly the average salary.

Observe the output.

```sql
select * from employees
where
salary=(select avg(salary) from employees);
```

Result Grid | Filter Rows: | Edit:

id	name	salary	department_id
NULL	NULL	NULL	NULL

7.

Display employees earning greater than the average salary but less than the highest salary.

```sql
select * from employees
where
salary>(select avg(salary) from employees) and salary < (select max(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name	salary	department_id
101	Dinesh	80000.00	1
104	Anitha	75000.00	2
105	Kalyan	85000.00	3
108	Siddhu	81000.00	2
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

8.

Display employees earning less than the company average but greater than the minimum salary.

```sql
select * from employees
where
salary < (select avg(salary) from employees) and salary > (select min(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name	salary	department_id
102	Akhila	68000.00	2
103	Sushma	65000.00	1
107	Vignesh	40000.00	4
NULL	NULL	NULL	NULL

9.

Find employees earning exactly the maximum salary.

```sql
select * from employees
where
salary=(select max(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import

id	name	salary	department_id
109	Bob	90000.00	1
NULL	NULL	NULL	NULL

10.

Find employees earning exactly the minimum salary.

```sql
select * from employees
where
salary=(select min(salary) from employees);
```

Result Grid | Filter Rows: | Edit:

id	name	salary	department_id
106	Babu	35000.00	4
NULL	NULL	NULL	NULL

Section B – Multi-row Subqueries

Use **only Multi-row Subqueries** with the **IN** operator.

11.

Display employees working in Hyderabad departments.

```sql
select * from employees
where
department_id in (select id from departments where location='Hyderabad');
```

id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
105	Kalyan	85000.00	3
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

12.

Display employees working in Bangalore departments.

```sql
select * from employees
where
department_id in (select id from departments where location = 'Bangalore');
```

id	name	salary	department_id
102	Akhila	68000.00	2
104	Anitha	75000.00	2
108	Siddhu	81000.00	2
NULL	NULL	NULL	NULL

13.

Display employees working in Chennai departments.

```sql
select * from employees
where
department_id = (select id from departments where location = 'Chennai');
```

id	name	salary	department_id
106	Babu	35000.00	4
107	Vignesh	40000.00	4
NULL	NULL	NULL	NULL

14.

Display employees working in Engineering or Sales departments.

```
select * from employees
where
department_id in (select id from departments
where
department_name='Engineering' or department_name='Sales');
```

Result Grid | Filter Rows: | Edit: | Export/Import:

id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
102	Akhila	68000.00	2
104	Anitha	75000.00	2
108	Siddhu	81000.00	2
NULL	NULL	NULL	NULL

15.

Display employees belonging to departments whose ID is greater than 2.

```
select * from employees
where
department_id in (select id from departments where id > 2);
```

Result Grid | Filter Rows: | Edit: | Export/Import:

id	name	salary	department_id
105	Kalyan	85000.00	3
110	Pavan	82000.00	3
106	Babu	35000.00	4
107	Vignesh	40000.00	4
NULL	NULL	NULL	NULL

16.

Display employees belonging to departments located in Hyderabad or Chennai.

```sql
select * from employees
where
department_id in (select id from departments
where location ='Hyderabad' or location='Chennai');
```

Result Grid | Filter Rows: | Edit: | Expo

id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
105	Kalyan	85000.00	3
110	Pavan	82000.00	3
106	Babu	35000.00	4
107	Vignesh	40000.00	4
NULL	NULL	NULL	NULL

17.

Display employees working in departments whose names start with the letter **S**.

```sql
select * from employees
where
department_id in (select id from departments where department_name like 's%');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content

id	name	salary	department_id
102	Akhila	68000.00	2
104	Anitha	75000.00	2
108	Siddhu	81000.00	2
NULL	NULL	NULL	NULL

18.

Display employees working in departments located outside Bangalore.

```sql
select * from employees
where
department_id in (select id from departments where
location != 'Bangalore');
```

id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
105	Kalyan	85000.00	3
110	Pavan	82000.00	3
106	Babu	35000.00	4
107	Vignesh	40000.00	4
NULL	NULL	NULL	NULL

19.

Display employees belonging to departments with IDs less than 3.

```sql
select * from employees
where
department_id in (select id from departments where id <3);
```

id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
102	Akhila	68000.00	2
104	Anitha	75000.00	2
108	Siddhu	81000.00	2
NULL	NULL	NULL	NULL

20.

Display employees belonging to all departments located in Hyderabad.

```sql
select * from employees
where
department_id in (select id from departments where location = 'Hyderabad');
```

id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
105	Kalyan	85000.00	3
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

Challenge Question

Try replacing **IN** with **=** in one of the above queries.

Observe the error and explain why it occurs.

0 19 17:49:35	select * from employees where department_id = (select id from departments where location = 'Hyderabad') LIMIT 0, 1000	Error Code: 1242. Subquery returns more than 1 row	0.000 sec

---

Section C – Correlated Subqueries

Use **only Correlated Subqueries**.

21.

Display employees earning more than the average salary of their own department.

```sql
select * from employees e
where
salary > (select avg(salary) from employees
where department_id = e.department_id);
```

id	name	salary	department_id
101	Dinesh	80000.00	1
104	Anitha	75000.00	2
105	Kalyan	85000.00	3
107	Vignesh	40000.00	4
108	Siddhu	81000.00	2
109	Bob	90000.00	1
NULL	NULL	NULL	NULL

22.

Display the highest paid employee from each department.

```sql
select * from employees e
where
salary = (select max(salary) from employees where department_id = e.department_id);
```

id	name	salary	department_id
105	Kalyan	85000.00	3
107	Vignesh	40000.00	4
108	Siddhu	81000.00	2
109	Bob	90000.00	1
NULL	NULL	NULL	NULL

23.

Display the lowest paid employee from each department.

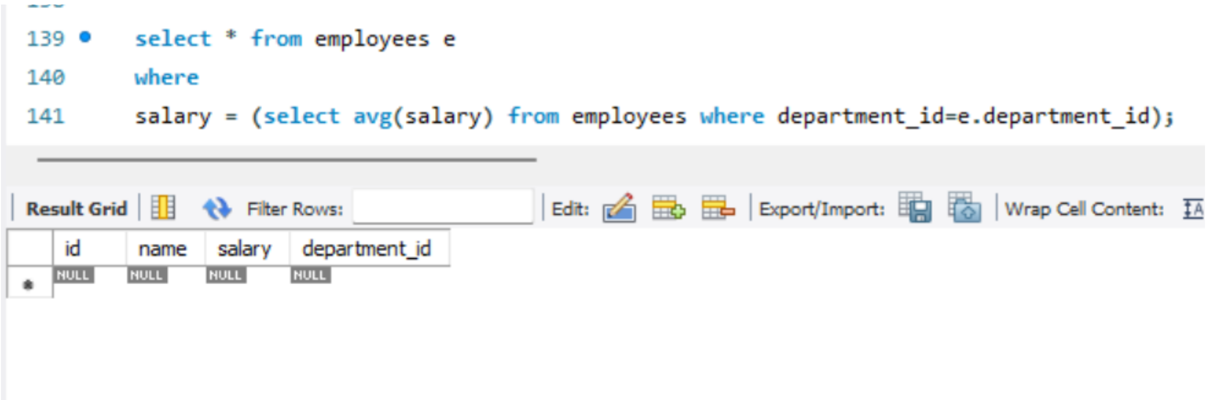
```sql
select * from employees e
where
salary =(select min(salary) from employees where department_id=e.department_id);
```

id	name	salary	department_id
102	Akhila	68000.00	2
103	Sushma	65000.00	1
106	Babu	35000.00	4
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

24.

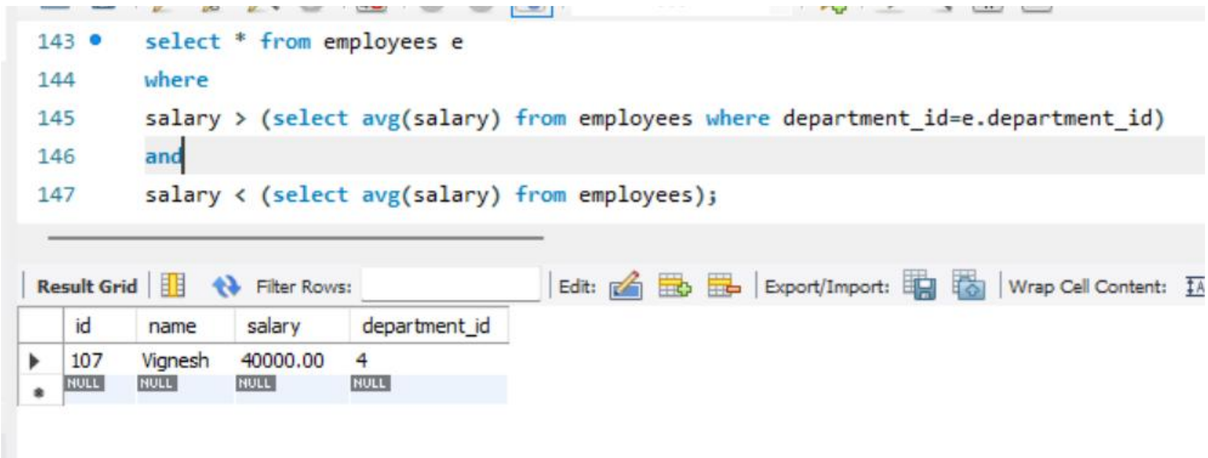
Display employees earning exactly the average salary of their department.

Observe the output.



25.

Display employees earning above their department average but below the company average.



26.

Display employees earning below their department average.

```sql
select * from employees e
where
salary < (select avg(salary) from employees where department_id=e.department_id);
```

id	name	salary	department_id
102	Akhila	68000.00	2
103	Sushma	65000.00	1
106	Babu	35000.00	4
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

27.

Display employees whose salary is equal to the maximum salary in their department.

```sql
select * from employees e
where
salary = (select max(salary) from employees where department_id=e.department_id);
```

id	name	salary	department_id
105	Kalyan	85000.00	3
107	Vignesh	40000.00	4
108	Siddhu	81000.00	2
109	Bob	90000.00	1
NULL	NULL	NULL	NULL

28.

Display employees whose salary is equal to the minimum salary in their department.

```sql
select * from employees e
where
salary = (select min(salary) from employees where department_id = e.department_id);
```

id	name	salary	department_id
102	Akhila	68000.00	2
103	Sushma	65000.00	1
106	Babu	35000.00	4
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL

29.

Display employees whose salary is greater than every other employee in their own department.

```sql
select * from employees e
where
salary = (select max(salary) from employees where department_id =  e.department_id);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name	salary	department_id
105	Kalyan	85000.00	3
107	Vignesh	40000.00	4
108	Siddhu	81000.00	2
109	Bob	90000.00	1
NULL	NULL	NULL	NULL

30.

Display employees whose salary is less than every other employee in their own department.

```sql
select * from employees e
where
salary = (select min(salary) from employees where department_id = e.department_id);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name	salary	department_id
102	Akhila	68000.00	2
103	Sushma	65000.00	1
106	Babu	35000.00	4
110	Pavan	82000.00	3
NULL	NULL	NULL	NULL